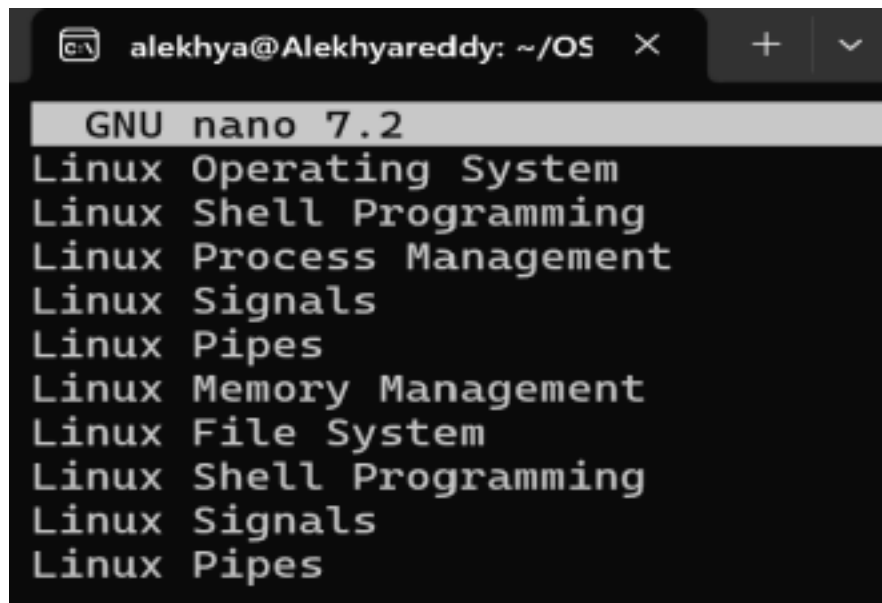


SKILL WEEK 12

1 .

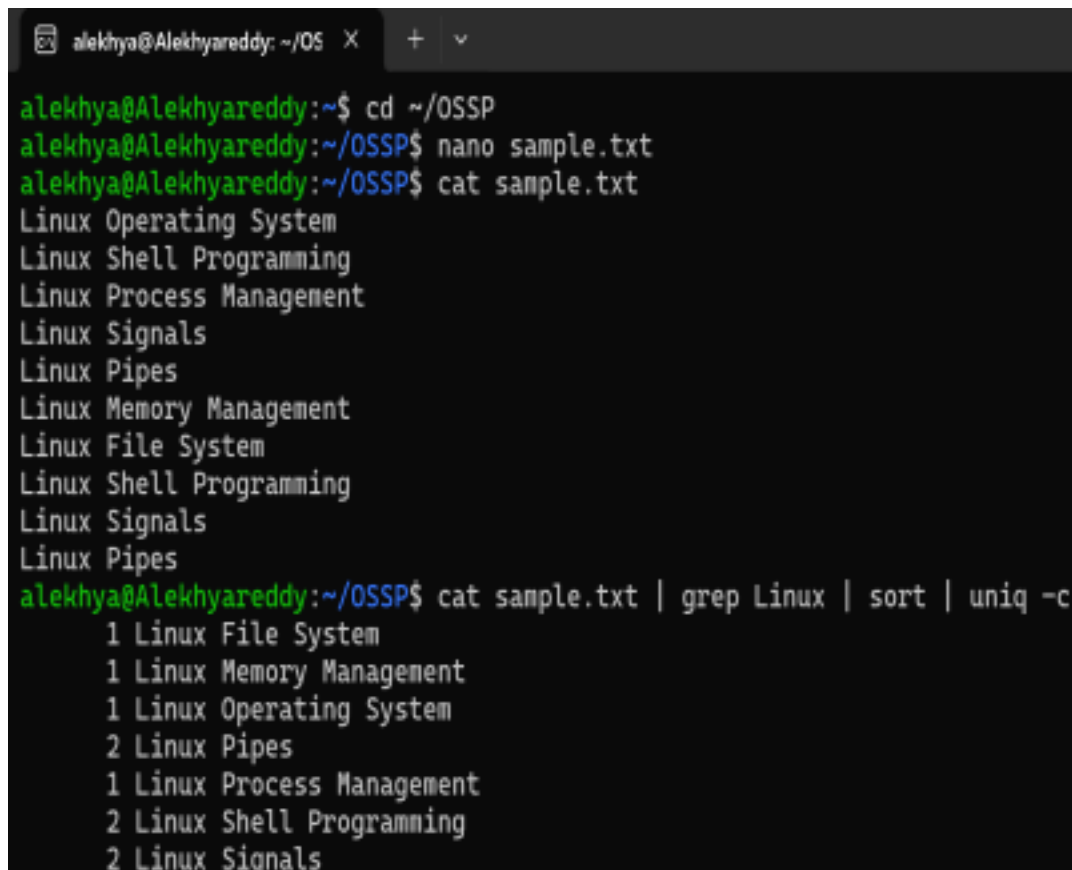
Aim: To execute Linux pipelines, launch multiple background jobs, monitor running processes and system resources, detect job failures, and analyze performance using Linux commands.

Program :



```
alekhya@Alekhyareddy: ~/OS X + v
GNU nano 7.2
Linux Operating System
Linux Shell Programming
Linux Process Management
Linux Signals
Linux Pipes
Linux Memory Management
Linux File System
Linux Shell Programming
Linux Signals
Linux Pipes
```

Output:



```
alekhya@Alekhyareddy:~$ cd ~/OSSP
alekhya@Alekhyareddy:~/OSSP$ nano sample.txt
alekhya@Alekhyareddy:~/OSSP$ cat sample.txt
Linux Operating System
Linux Shell Programming
Linux Process Management
Linux Signals
Linux Pipes
Linux Memory Management
Linux File System
Linux Shell Programming
Linux Signals
Linux Pipes
alekhya@Alekhyareddy:~/OSSP$ cat sample.txt | grep Linux | sort | uniq -c
  1 Linux File System
  1 Linux Memory Management
  1 Linux Operating System
  2 Linux Pipes
  1 Linux Process Management
  2 Linux Shell Programming
  2 Linux Signals
```

alekhya@Alekhyareddy: ~/OS X + v

alekhya@Alekhyareddy:~/OSSP\$ sleep 200 &

[1] 469

alekhya@Alekhyareddy:~/OSSP\$ sleep 250 &

[2] 473

alekhya@Alekhyareddy:~/OSSP\$ sleep 300 &

[3] 474

alekhya@Alekhyareddy:~/OSSP\$ jobs -l

[1] 469 Running sleep 200 &

[2]- 473 Running sleep 250 &

[3]+ 474 Running sleep 300 &

alekhya@Alekhyareddy:~/OSSP\$ ps -f

UID	PID	PPID	C	STIME	TTY	TIME	CMD
alekhya	302	301	0	05:27	pts/0	00:00:00	-bash
alekhya	469	302	0	05:33	pts/0	00:00:00	sleep 200
alekhya	473	302	0	05:33	pts/0	00:00:00	sleep 250
alekhya	474	302	0	05:33	pts/0	00:00:00	sleep 300
alekhya	475	302	0	05:33	pts/0	00:00:00	ps -f

alekhya@Alekhyareddy:~/OSSP\$ top

top - 05:34:07 up 7 min, 1 user, load average: 0.06, 0.02, 0.00

Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie

%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st

MiB Mem : 15743.6 total, 14031.9 free, 571.1 used, 1337.1 buff/cache

MiB Swap: 4096.0 total, 4096.0 free, 0.0 used. 15172.5 avail Mem

PID to signal/kill [default pid = 1]

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	21740	13156	9800	S	0.0	0.1	0:00.43	systemd
2	root	20	0	3180	2208	2072	S	0.0	0.0	0:00.00	init-systemd(Ub
6	root	20	0	3196	2128	2008	S	0.0	0.0	0:00.00	init
52	root	19	-1	50368	16240	15112	S	0.0	0.1	0:00.13	systemd-journal
102	root	20	0	25756	6920	5168	S	0.0	0.0	0:00.13	systemd-udevd
130	systemd+	20	0	21344	13208	11088	S	0.0	0.1	0:00.06	systemd-resolve
131	systemd+	20	0	91036	7972	7008	S	0.0	0.0	0:00.02	systemd-timesyn
167	root	20	0	4244	2684	2432	S	0.0	0.0	0:00.01	cron
168	message+	20	0	9608	5224	4540	S	0.0	0.0	0:00.03	dbus-daemon
182	root	20	0	17976	8884	7848	S	0.0	0.1	0:00.05	systemd-logind
188	root	20	0	1756628	13572	11444	S	0.0	0.1	0:00.14	wsl-pro-service
204	syslog	20	0	222516	6024	4628	S	0.0	0.0	0:00.05	rsyslogd
207	root	20	0	3124	1972	1836	S	0.0	0.0	0:00.00	agetty
220	root	20	0	107036	23108	13656	S	0.0	0.1	0:00.14	unattended-upgr
300	root	20	0	3184	1112	980	S	0.0	0.0	0:00.00	SessionLeader
301	root	20	0	3200	1248	1108	S	0.0	0.0	0:00.02	Relay(302)
302	alekhya	20	0	6208	5480	3708	S	0.0	0.0	0:00.04	bash
303	root	20	0	6700	4608	3828	S	0.0	0.0	0:00.00	login

```

alekhy@AlekhyaReddy: ~/OSSP$ kill %2
alekhy@AlekhyaReddy: ~/OSSP$ jobs -l
[1] 469 Running          sleep 200 &
[2] 473 Terminated     sleep 250
[3]- 474 Running        sleep 300 &
[4]+ 476 Stopped (signal) top
alekhy@AlekhyaReddy: ~/OSSP$ time ls -R
.:
LICENSE                dynamic_memory          memory_leak             task1_wait.c           task5_process           task_valgrind.c
Makefile               dynamic_memory.c        memory_leak.c           task1_wait.c.save      task5_process.c         task_valgrind_leak
README.md              error.log               mmap_file               task2_quotes           task_error_handling     task_valgrind_leak.c
assets                 expected_report.txt     mmap_file.c             task2_quotes.c         task_error_handling.c   test_report.txt
bin                    include                 mmap_file.txt           task2_single_quotes    task_large_pipeline     test_script.sh
calculator.c            input.txt               module_test             task2_single_quotes.c  task_large_pipeline.c   tests
calculator.h            io_redirect             obj                     task3_double_quotes    task_process_groups     thread_counter
calculator.o            io_redirect.c           output.txt              task3_double_quotes.c  task_process_groups.c   thread_counter.c
calculator_test         library_copy            producer_consumer       task3_fork             task_resume_job         thread_counter_mutex
calculator_test.c       library_copy.c          producer_consumer.c     task3_fork.c           task_resume_job.c       thread_counter_mutex.c
client                 library_copy.txt        sample.txt              task3_input            task_sigint_forward     traditional_io
client.c               lowlevel_copy           scripts                  task3_input.c          task_sigint_forward.c   traditional_io.c
client_to_server        lowlevel_copy.c         server                   task4_escape           task_sigint_forward.c.save valgrind_report.txt
cow_practical           lowlevel_copy.txt       server.c                task4_escape.c         task_signal_actions     week4_T2
cow_practical.c         ls_grep                 server_to_client        task4_history           task_signal_actions.c   week4_T2.c
deadlock               ls_grep.c              signal_handler           task4_history.c        task_signal_behavior     week5_double_quotes.c
deadlock.c             main_loop               signal_handler.c        task4_states           task_signal_behavior.c   week5_double_quotes
deadlock_prevention    main_loop.c             source.txt              task4_states.c         task_sigsttp            task_valgrind
deadlock_prevention.c  main_module.c           src                     task5_memory           task_sigsttp.c
docs                   main_module.o           task1_wait              task5_memory.c         task_valgrind

./assets:
architecture.png

./bin:
fork_exec

./docs:
architecture.txt design.md design_decisions.txt execution_flow.txt report.pdf syscalls.txt

./include:
builtin.h executor.h parser.h shell.h

./obj:
builtin.o executor.o main.o parser.o shell.o

./scripts:
run.sh

./src:
builtin.c executor.c fork_exec.c main.c parser.c shell.c

./tests:

```

2.

Aim: To document the architecture of a Linux program, explain its execution path and system calls, create a flow diagram, record design decisions, and review the program documentation.

Program:

```
GNU nano 7.2
#include <stdio.h>
#include <unistd.h>
#include <time.h>
#include <sys/wait.h>

int main()
{
    printf("Program Started\n");

    printf("Process ID : %d\n", getpid());
    printf("Parent Process ID : %d\n", getppid());

    printf("Executing Program...\n");

    sleep(2);

    printf("Program Finished Successfully\n");

    return 0;
}
```

Output:

[illegible]

```

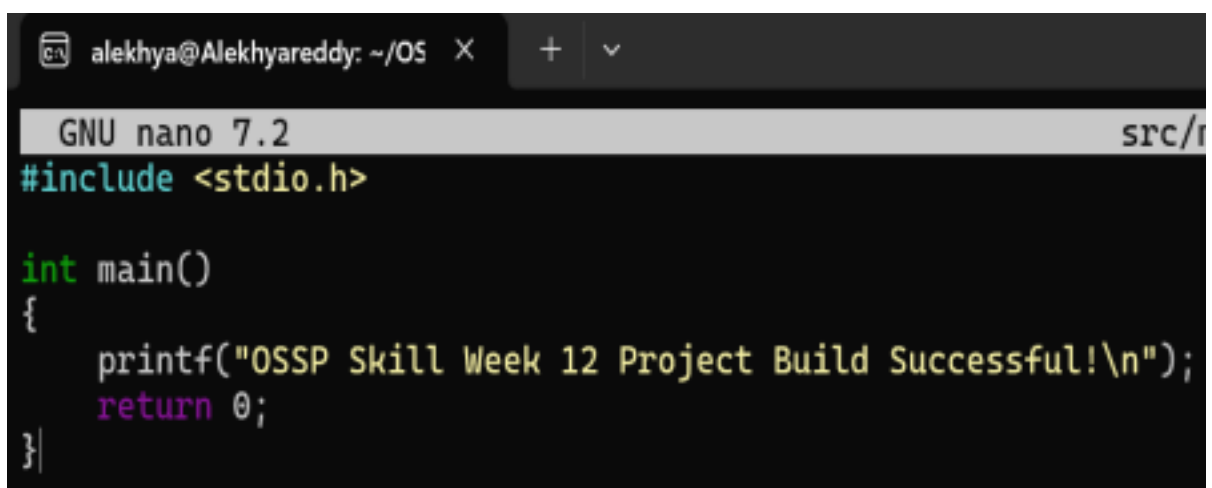
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fdb3813a000
arch_prctl(ARCH_SET_FS, 0x7fdb3813a740) = 0
set_tid_address(0x7fdb3813aa10) = 544
set_robust_list(0x7fdb3813aa20, 24) = 0
rseq(0x7fdb3813b060, 0x20, 0, 0x53053053) = 0
mprotect(0x7fdb37fff000, 16384, PROT_READ) = 0
mprotect(0x6412af794000, 4096, PROT_READ) = 0
mprotect(0x7fdb38185000, 8192, PROT_READ) = 0
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) = 0
munmap(0x7fdb3813d000, 32539) = 0
fstat(1, {st_mode=S_IFCHR|0620, st_rdev=nakedev(0x88, 0), ...}) = 0
getrandom("\xa5\x02\x66\xb7\xff\x21\xa2\xcc", 8, GRND_NONBLOCK) = 8
brk(NULL) = 0x6412d3eaf000
brk(0x6412d3ed0000) = 0x6412d3ed0000
write(1, "Program Started\n", 16Program Started
) = 16
getpid() = 544
write(1, "Process ID : 544\n", 17Process ID : 544
) = 17
getppid() = 541
write(1, "Parent Process ID : 541\n", 24Parent Process ID : 541
) = 24
write(1, "Executing Program...\n", 21Executing Program...
) = 21
clock_nanosleep(CLOCK_REALTIME, 0, {tv_sec=2, tv_nsec=0}, 0x7ffeb9088e90) = 0
write(1, "Program Finished Successfully\n", 30Program Finished Successfully
) = 30
exit_group(0) = ?
+++ exited with 0 +++

```

3.

Aim: To organize source code into separate folders, create a build script, add a README file, verify the repository structure, and validate the project deliverables in Linux.

Program:



```

GNU nano 7.2 src/i
#include <stdio.h>

int main()
{
    printf("OSSP Skill Week 12 Project Build Successful!\n");
    return 0;
}

```

```
alekhya@Alekhyareddy: ~/OS × +
GNU nano 7.2
#ifdef CALCULATOR_H
#define CALCULATOR_H

int add(int a, int b);
int subtract(int a, int b);

#endif|
```

```
alekhya@Alekhyareddy: ~/OS × + v
GNU nano 7.2 docs/README.md
# OSSP Skill Week 12

## Project
Operating System and Systems Programming

## Description
This project demonstrates Linux system programming concepts using C.

## Build
The project is compiled using GCC.

## Run
The generated executable is used to test the project.

## Structure
src      - Source files
include  - Header files
docs     - Documentation|
```

```
alekhya@Alekhyareddy: ~/OS × + v
GNU nano 7.2
#!/bin/bash

echo "Building OSSP Project..."

gcc src/main.c -o project_build

if [ $? -eq 0 ]; then
    echo "Build Completed Successfully."
else
    echo "Build Failed."
fi|
```

```
alekhya@Alekhyareddy: ~/OS × + ▾
GNU nano 7.2
OSSP SKILL WEEK 12 REPORT

Project Build and Repository Organization

Source Directory:
src/

Header Directory:
include/

Documentation Directory:
docs/

Build Script:
build.sh

Executable:
project_build

Build Status:
Successful|
```

Output:

alekhy@AlekhyaReddy: ~/OS X + v

alekhy@AlekhyaReddy:~\$ cd ~/OSSP

alekhy@AlekhyaReddy:~/OSSP\$ mkdir -p src include docs

alekhy@AlekhyaReddy:~/OSSP\$ touch src/main.c

alekhy@AlekhyaReddy:~/OSSP\$ touch include/calculator.h

alekhy@AlekhyaReddy:~/OSSP\$ touch docs/README.md

alekhy@AlekhyaReddy:~/OSSP\$ ls -R

..

LICENSE	include	producer_consumer	task3_input	task_signal_actions.c
Makefile	input.txt	producer_consumer.c	task3_input.c	task_signal_behavior
README.md	io_redirect	sample.txt	task4_escape	task_signal_behavior.c
assets	io_redirect.c	scripts	task4_escape.c	task_sigstsp
bin	library_copy	server	task4_history	task_sigstsp.c
calculator.c	library_copy.c	server.c	task4_history.c	task_valgrind
calculator.h	library_copy.txt	server_to_client	task4_states	task_valgrind.c
calculator.o	lowlevel_copy	signal_handler	task4_states.c	task_valgrind_leak
calculator_test	lowlevel_copy.c	signal_handler.c	task5_memory	task_valgrind_leak.c
calculator_test.c	lowlevel_copy.txt	skill12_task2	task5_memory.c	test_report.txt
client	ls_grep	skill12_task2.c	task5_process	test_script.sh
client.c	ls_grep.c	source.txt	task5_process.c	tests
client_to_server	main_loop	src	task_error_handling	thread_counter
cow_practical	main_loop.c	task1_wait	task_error_handling.c	thread_counter.c
cow_practical.c	main_module.c	task1_wait.c	task_large_pipeline	thread_counter_mutex
deadlock	main_module.o	task1_wait.c.save	task_large_pipeline.c	thread_counter_mutex.c
deadlock.c	memory_leak	task2_quotes	task_process_groups	traditional_io
deadlock_prevention	memory_leak.c	task2_quotes.c	task_process_groups.c	traditional_io.c
deadlock_prevention.c	mmap_file	task2_single_quotes	task_resume_job	valgrind_report.txt
docs	mmap_file.c	task2_single_quotes.c	task_resume_job.c	week4_T2
dynamic_memory	mmap_file.txt	task3_double_quotes	task_sigint_forward	week4_T2.c
dynamic_memory.c	module_test	task3_double_quotes.c	task_sigint_forward.c	week_5_double_quotes.c
error.log	obj	task3_fork	task_sigint_forward.c.save	week_5double_quotes
expected_report.txt	output.txt	task3_fork.c	task_signal_actions	

```
./tests:
test_executor.c test_parser.c
alekhya@Alekhyareddy:~/OSSP$ nano src/main.c
alekhya@Alekhyareddy:~/OSSP$ nano include/calculator.h
alekhya@Alekhyareddy:~/OSSP$ nano docs/README.md
alekhya@Alekhyareddy:~/OSSP$ nano build.sh
alekhya@Alekhyareddy:~/OSSP$ chmod +x build.sh
alekhya@Alekhyareddy:~/OSSP$ ./build.sh
Building OSSP Project...
Build Completed Successfully.
alekhya@Alekhyareddy:~/OSSP$ ./project_build
OSSP Skill Week 12 Project Build Successful!
alekhya@Alekhyareddy:~/OSSP$ nano report.txt
alekhya@Alekhyareddy:~/OSSP$ tree
```

```
.
├── LICENSE
├── Makefile
├── README.md
├── assets
│   └── architecture.png
├── bin
│   └── fork_exec
├── build.sh
├── calculator.c
├── calculator.h
├── calculator.o
├── calculator_test
├── calculator_test.c
├── client
├── client.c
├── client_to_server
├── cow_practical
├── cow_practical.c
├── deadlock
├── deadlock.c
├── deadlock_prevention
├── deadlock_prevention.c
├── docs
│   ├── README.md
│   ├── architecture.txt
│   ├── design.md
│   ├── design_decisions.txt
│   ├── execution_flow.txt
│   ├── report.pdf
│   └── syscalls.txt
├── dynamic_memory
├── dynamic_memory.c
├── error.log
├── expected_report.txt
├── include
│   └── builtin.h
```



alekhya@Alekhyareddy: ~/OS



```
— task4_escape.c
— task4_history
— task4_history.c
— task4_states
— task4_states.c
— task5_memory
— task5_memory.c
— task5_process
— task5_process.c
— task_error_handling
— task_error_handling.c
— task_large_pipeline
— task_large_pipeline.c
— task_process_groups
— task_process_groups.c
— task_resume_job
— task_resume_job.c
— task_sigint_forward
— task_sigint_forward.c
— task_sigint_forward.c.save
— task_signal_actions
— task_signal_actions.c
— task_signal_behavior
— task_signal_behavior.c
— task_sigtstp
— task_sigtstp.c
— task_valgrind
— task_valgrind.c
— task_valgrind_leak
— task_valgrind_leak.c
— test_report.txt
— test_script.sh
— tests
  |— test_executor.c
  |— test_parser.c
— thread_counter
— thread_counter.c
— thread_counter_mutex
— thread_counter_mutex.c
— traditional_io
— traditional_io.c
— valgrind_report.txt
— week4_T2
— week4_T2.c
— week_5_double_quotes.c
— week_5double_quotes
```

9 directories, 142 files

alekhya@Alekhyareddy:~/OSSP\$ |

Aim: To create demo scenarios, validate Linux features, demonstrate pipelines and signals, verify program execution, and perform a final project review in Linux.

Program :

```
alekhya@Alekhyareddy: ~/OS ×
GNU nano 7.2
Linux Memory
Linux Operating System
Linux Pipes
Linux Processes
Linux Shell
Linux Signals
Linux Signals
Linux Pipes|
```

Output:

```
alekhya@Alekhyareddy: ~/OS × + v
alekhya@Alekhyareddy:~$ cd ~/OSSP
alekhya@Alekhyareddy:~/OSSP$ nano demo.txt
alekhya@Alekhyareddy:~/OSSP$ cat demo.txt | grep Linux | sort | uniq -c
  1 Linux Memory
  1 Linux Operating System
  2 Linux Pipes
  1 Linux Processes
  1 Linux Shell
  2 Linux Signals
alekhya@Alekhyareddy:~/OSSP$ sleep 120 &
[1] 682
alekhya@Alekhyareddy:~/OSSP$ jobs -l
[1]+  682 Running                  sleep 120 &
alekhya@Alekhyareddy:~/OSSP$ kill -STOP PID
-bash: kill: PID: arguments must be process or job IDs
alekhya@Alekhyareddy:~/OSSP$ kill -STOP 7212
-bash: kill: (7212) - No such process
alekhya@Alekhyareddy:~/OSSP$ jobs -l
[1]+  682 Running                  sleep 120 &
alekhya@Alekhyareddy:~/OSSP$ jobs -l
[1]+  682 Running                  sleep 120 &
alekhya@Alekhyareddy:~/OSSP$ kill %1
alekhya@Alekhyareddy:~/OSSP$ jobs -l
[1]+  682 Terminated              sleep 120
```

```

alekhy@AlekhyaReddy: ~/OS X + v
[1]+  682 Terminated                  sleep 120
alekhy@AlekhyaReddy:~/OSSP$ ls -R
.:
LICENSE                expected_report.txt  producer_consumer    task3_fork.c          task_signal_actions.c
Makefile               include             producer_consumer.c  task3_input           task_signal_behavior.c
README.md              input.txt           project_build        task3_input.c         task_signal_behavior.c
assets                 io_redirect         report.txt           task4_escape          task_sigtstp
bin                    io_redirect.c      sample.txt          task4_escape.c        task_sigtstp.c
build.sh               library_copy       scripts             task4_history         task_valgrind
calculator.c           library_copy.c     server              task4_history.c       task_valgrind.c
calculator.h           library_copy.txt   server.c            task4_states          task_valgrind_leak
calculator.o           lowlevel_copy     server_to_client    task4_states.c        task_valgrind_leak.c
calculator_test        lowlevel_copy.c   signal_handler.c    task5_memory          test_report.txt
calculator_test.c      lowlevel_copy.txt signal_handler.c     task5_memory.c        test_script.sh
client                 ls_grep           skill12_task2       task5_process         tests
client.c              ls_grep.c         source.txt          task_error_handling   thread_counter
client_to_server       main_loop         src                 task_error_handling.c thread_counter.c
cow_practical          main_module.c     task1_wait          task_large_pipeline   thread_counter_mutex
cow_practical.c       main_module.o     task1_wait.c        task_large_pipeline.c traditional_io
deadlock              memory_leak        task1_wait.c.save   task_process_groups   traditional_io.c
deadlock_prevention   mmap_file          task2_quotes        task_process_groups.c valgrind_report.txt
deadlock_prevention.c mmap_file.c        task2_single_quotes task_resume_job        week4_T2
demo.txt              mmap_file.txt      task3_double_quotes task_sigint_forward    week4_T2.c
docs                  module_test        task3_double_quotes task_sigint_forward.c  week_5_double_quotes.c
dynamic_memory         obj               task3_double_quotes task_sigint_forward.c.save
dynamic_memory.c       output.txt         task3_fork           task_signal_actions
error.log             

./assets:
architecture.png

./bin:
fork_exec

./docs:
README.md  architecture.txt  design.md  design_decisions.txt  execution_flow.txt  report.pdf  syscalls.txt

./include:
builtin.h  calculator.h  executor.h  parser.h  shell.h

./obj:
builtin.o  executor.o  main.o  parser.o  shell.o

```

```

./assets:
architecture.png

./bin:
fork_exec

./docs:
README.md  architecture.txt  design.md  design_decisions.txt  execution_flow.txt  report.pdf  syscalls.txt

./include:
builtin.h  calculator.h  executor.h  parser.h  shell.h

./obj:
builtin.o  executor.o  main.o  parser.o  shell.o

./scripts:
run.sh

./src:
builtin.c  executor.c  fork_exec.c  main.c  parser.c  shell.c

./tests:
test_executor.c  test_parser.c
alekhy@AlekhyaReddy:~/OSSP$

```